Quantifying Mutual Intelligibility using Language Models: Evidence from Turkic Languages

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Mutual intelligibility (MI) is the ability of speakers of one language to understand speakers of another, often due to overlapping lexicon and sound patterns. Existing computational approaches quantify MI using lexical or phonetic distances computed over hand-curated cognate or high-frequency word lists in related languages, and they predominantly focus on Indo-European languages [1]. In this study, we investigate MI within the understudied Turkic language family using neural language modeling (LM). Turkic languages are typologically similar, sharing phonetic inventories and phonological processes such as vowel harmony [2,3], as well as substantial lexical overlap within subgroups [4]. We speculate that shared phonetic and phonological structure, along with lexical similarity within Turkic subgroups, facilitates recognition and discrimination of related-language input. This hypothesis is supported by prior work showing that non-native phonology can hinder comprehension more than morphosyntactic differences, highlighting the importance of phonetic similarity alongside lexical similarity in related languages [5]. Unlike list-based approaches, our method derives similarity directly from model behavior on continuous text, without curated cognate sets or pairwise comparisons.

Approach: We hypothesize that an LM trained on a source language and fine-tuned on more closely related target languages will exhibit greater generalizability than when fine-tuned on distantly related ones. We operationalize this as the model's character-level cross-entropy (CE) loss, an aggregate measure of surprise over phoneme sequences. We predict that training and test CE loss will reflect model convergence and generalization to the target language, capturing phonological and lexical predictability beyond curated cognate lists. To approximate pronunciation across Turkic languages, we transcribe all texts into IPA and train a character-level LSTM, avoiding script inconsistency [6] and tokenizer design issues in highly agglutinative, low-resource languages [7] (Table 1).

Methods: We extracted raw text for closely related Kazakh-Kyrgyz, Turkish-Azerbaijani, and Uzbek-Uyghur language pairs from the OSCAR web corpus, filtered it with FastText language identification (≥ 0.95 confidence) to reduce contamination, and then deterministically transliterated all data into broad IPA using language-specific rule sets from the phonological literature [8, 9]. We trained a 2-layer character-level LSTM (128-dimensional embeddings, 256 hidden units) [10] in a sequential bilingual regimen [11], first training and validating on the source languages Kazakh, Uzbek, and Turkish and fine-tuning on the closely related target languages Kyrgyz, Uyghur, and Azerbaijani along one distant language per pair. We also added a distractor language to the model – Finnish, which is known to be typologically close (agglutinative with vowel harmony) but genealogically unrelated.

Results: Across all experiments, CE patterns aligned with known MI gradients: the Kazakh model showed a smaller spike in loss, faster convergence, and lower final test loss on Kyrgyz than on Turkish, and likewise the Turkish model transferred more efficiently to Azerbaijani than to Kazakh, while the Uzbek model transferred at least slightly better to Uyghur than to Kazakh (Figure 1). Although absolute differences in test loss were numerically small, they consistently preserved the same ordering as known MI ratings (Kyrgyz > Turkish for Kazakh, Azerbaijani > Kazakh for Turkish, Uyghur > Kazakh for Uzbek), indicating that even subtle variation in CE loss is sensitive to MI gradients between Turkic language pairs (Figure 2). Finnish showed higher CE loss for all language pairs.

Conclusion: Our results suggest that a character-level LM trained on a source language finds closely related varieties more plausible than more distant ones, as reflected in lower CE loss and faster convergence, thereby mirroring known MI gradients within the family. The lower performance on Finnish, suggests that phonological overlap alone is insufficient, and genealogical

relatedness appears to play a central role in shaping linguistic transfer. Training on additional language pairs is in progress, and we plan to collect native-speaker MI judgments for the same language pairs to directly test how well model-based CE differences predict human MI ratings.

Language	Branch	Sentence in original script	IPA transcription
Turkish Azerbaijani	Oghuz	(Ben) (bir) kitap okuyorum. (Mən) kitab oxuyuram.	(ben bɨɾ) kitap okujoɾum. (mæn) kitɑb oxujuɾɑm.
Kazakh Kyrgyz	Kipchak	(Мен) кітап оқып жатырмын. (Мен) китеп окуп жатам.	(men) kɪtɑp oqəp ʒɑtəɾmən. (men) kitep okup ʒɑtɑm.
Uzbek Uyghur	Karluk	(Men) kitob o'qiyapman. (Men) kitab oquwatimen*	(men) kitɔb oqijapman. (mæn) kitɑb oquwɑtimæn.

Table 1: MI among Turkic languages illustrated by the sentence: *I am reading a book.* *Latin script was used for Uyghur because the Arabic script caused compilation errors.

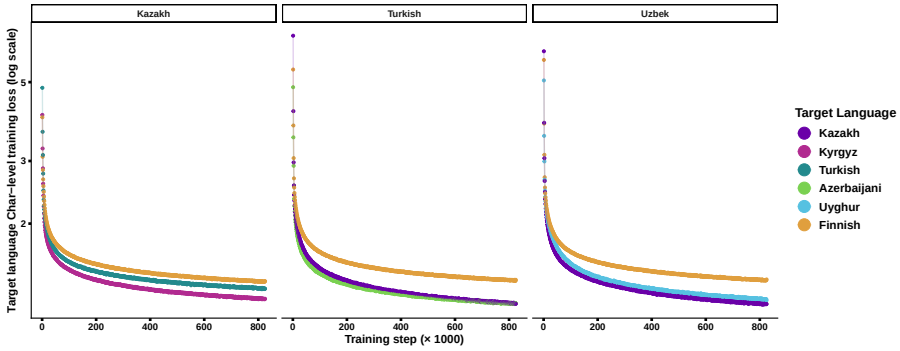

Figure 1: Character-level CE loss across training steps; models trained on Kazakh, Turkish, and Uzbek, then fine-tuned on Kyrgyz, Azerbaijani, Uyghur and Finnish.

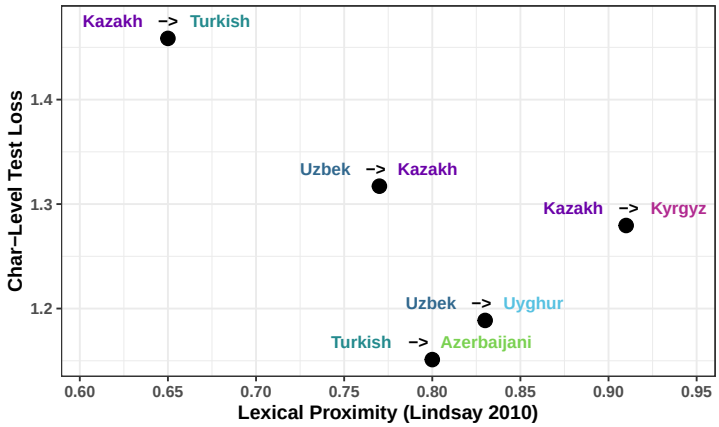

Figure 2: Character-level test loss plotted against lexical proximity, defined by cognate-based MI estimates. R=-.67

Selected References:[1] Gooskens & Heuven (2021).*Mutual Intelligibility.* [2] Baskakov (1988). *Istoriko-tipologicheskaya fonologiya tyurkskih yazykov.* [3] Tambovtsev (2001). *The phonological distances between Turkic languages based on some phonological features of consonants.* [4] Lindsay (2010). *Mutual intelligibility among the Turkic languages.* [5] Hilton et al. (2013). *The influence of non-native morphosyntax on the intelligibility of a closely related language.* [6] Mirzakhalov et al. (2021). *A large-scale study of machine translation in the Turkic languages.* [7] Toraman et al. (2023). *Impact of tokenization on language models: An analysis for Turkish.* [8] McCollum et al. (2021) Kazakh. [9] Zimmer et al. (1992) Turkish. [10] Hochreiter et al. (1997). *Long short-term memory.* [11] Arnett et al. (2025). *On the acquisition of shared grammatical representations in bilingual language models.*